Scholar Link: A Unified Platform for Scholarship and Higher Education Management

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***Abstract*—The digitalisation of information sources, such as in higher education has increased access to information but also has led to problems such as data isolation, out-of-date scholarship listings and a lack of support, especially for scholarship-reliant low-income countries. This paper proposes Scholar Link, a web platform that collates information on universities, programs of study, scholarships, mentors and students. Scholar Link provides scholarship recommendations, mentorship, deadline reminders, and reviews. Scholar Link offers trusted and tailored information which makes it more accessible, efficient, and trustworthy in the scholarship search process.**

I. Introduction

Equality in higher education is crucial for individual growth and national advancement through improved employment opportunities, innovation and social mobility. But access to higher education is not equally available, especially for low-income and developing country students. Scholarships and financial assistance programs are the key resources for many of these students to access higher education, particularly in overseas universities. In the absence of these, many talented students are unable to fulfill their educational aspirations due to financial barriers. The rapid development of information technologies has led students to increasingly rely on online platforms to find universities, courses and scholarships. Such websites have made information more accessible than in the past. But in spite of this, there are still challenges. Scholarship and program information is often spread across several websites, requiring students to spend hours searching for the right information. Moreover, the information is often incomplete, outdated or inconsistent, forcing students to cross-check information from multiple sources. Other than official platforms, students also use social media platforms like online forums and groups to find scholarships. While these may provide useful information, they can be disorganised and lack credibility, leading to confusion and inaccuracies. Furthermore, existing systems may not provide personalised advice, guidance and time lines, which are crucial to the application process. This can lead to missed opportunities or wrong decisions. The other major issue with current systems is a lack of trust. The majority of systems provide links to other websites without checking that they are trustworthy, and this may cause incorrect information. Lastly, there is little opportunity for users to give feedback and reviews, also reducing their trustworthiness. This project aims to develop Scholar Link, a website that seeks to bring together information on universities, higher education courses, scholarships, mentorship and students. The system aims to make the scholarship search and selections by providing students with accurate information, tailored suggestions, mentorship and time management. Through this integration, Scholar Link increases access to information, supports informed decision-making and provides a trusted and convenient platform for students looking for higher education opportunities.

II. Literature Review

Digital platforms increasingly play a role in enabling access to higher education, but there are still many problems with the lack of information integration, personalization, and integration. Prior work has examined different aspects of financial aid, recommender systems, student behaviour and digitalisation in education, and has identified areas of progress and improvement. Herbaut and Geven [1] present compelling evidence that financial aid contributes to enhancing access to higher education, particularly for underrepresented and disadvantaged students. They stress that grants and scholarships are essential to increasing educational equity. But their research is primarily concerned with the policy implications on a large scale and not how technology can support students in identifying, comparing and applying for such funding opportunities. In the field of recommender systems, Kamal et al. [2] explain how recommendation engines can improve educational decision making by alleviating information overload and recommending relevant opportunities. These systems show the possibilities of applying

machine learning and data mining techniques to education.But existing systems lack transparency in their algorithms, are based on imperfect or biased data sources, and don't consider practical constraints such as eligibility, applications deadlines and geographical locations for scholarships. Student behaviour studies also highlight the problems with academic support. Li et al. [3] show that students are not making full use of the academic support, largely due to lack of awareness, lack of accessibility, and lack of guidance. Their research focuses on behavioral challenges rather than technological aspects, demonstrating that even if resources are available, students may not be able to effectively use them. However, it does not offer a system-level approach that could provide structured guidance for students to access academic and financial aid opportunities. There has also been significant research on digital transformation in higher education. Mabotha and Ngcamu [4] examine the use of digital systems in education, which improve administrative efficiency, data storage and communication. However, they also reveal challenges such as the fragmentation of systems, interoperability problems across systems, and poor data management, which limit the effectiveness of digital transformation in education. Likewise, Serwornoo et al. [5] explore student engagement with digital academic platforms, and reveal multiple challenges such as inadequate data quality, system reliability, trust in online information and information overload. They conclude that while digital platforms are extensively used, they tend not to offer curated, validated and contextualised information, resulting in low user satisfaction and inefficiencies in decision-making. In summary, while the literature reviewed above shows that financial aid systems, recommender systems and digital education platforms have all made a contribution to increasing accessibility of higher education, we still have a long way to go in integrating these systems. Current solutions are largely independent and lack the integration of scholarship databases, information on academic programs, mentoring programs and personalised recommendation systems. Scholar Link bridges this gap by offering an integrated system that increases access, ensures reliability and enables decision-making by providing guidance and personalisation.

III. Problem Statement

The search for scholarships, universities and courses is uncoordinated. Students rely on social media and forums, but data is unfiltered, unreliable and outdated. Current systems lack filtering, mentorship and deadline tracking.

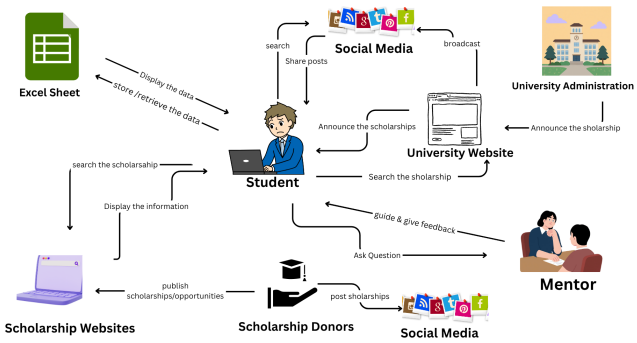


Fig. 1. Rich Picture AS-IS

The existing system, as depicted in the rich picture (AsIs), is a fragmented system where students, universities and scholarship providers are not integrated. These issues suggest the need for an integrated and reliable system such as Scholar Link to provide verified information, targeted recommendations, mentorship support and tracking.

TABLE I
Problem Analysis of Existing System

No	Process Name	Stakeholders	Concerns (Problems)	Root Cause Analysis	Proposed Solution
1	Information Discovery & Search	Prospective Students, Universities	Fragmented and outdated information across multiple platforms	Siloed databases and lack of API integration between systems	Centralized integrated hub aggregating verified data from primary sources
2	Eligibility & Profile Filtering	Students, Scholarship Providers	Irrelevant results and manual filtering required	Absence of structured student profiles and matching algorithm	Automated profile matching using relational database keys
3	Mentorship & Guidance	Applicants, Student Mentors	No structured guidance; reliance on external social media	Mentorship not integrated into system; no verified mentor database	Structured mentorship module with verified peer connections
4	Deadline & Opportunity Tracking	Students, Academic Advisors	Missed deadlines and outdated opportunities	Manual tracking without automated alerts or status updates	Automated deadline manager with notifications and active-status filtering
5	Data Verification & Trust	Students, Providers	Low trust due to unverified external data sources	No validation mechanism or verified data marking system	Verified data validation system with provider authentication portal
6	Institutional Communication	Universities, Admins	Poor communication of updates to applicants	Passive website updates without direct targeting	Provider dashboard enabling direct broadcast to eligible students

IV. Solution

Scholar Link is a system that combines all components into a single integrated platform.

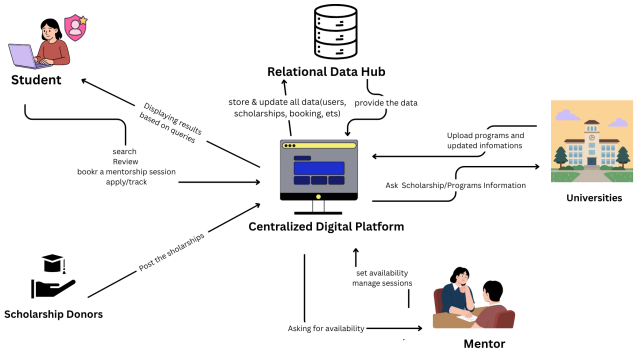


Fig. 2. Proposed Solution Rich Picture

The system integrates students, universities, mentors and scholarship providers into one platform.

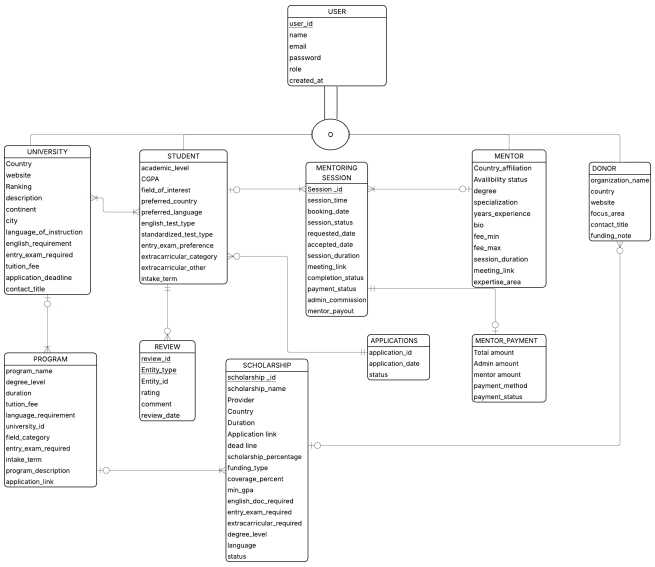


Fig. 3. Entity Relationship Diagram

The ERD defines relationships between User, Student, Mentor, University, Program, Scholarship, Application, Review and Mentoring Session.



Fig. 4. Database Schema

The schema defines structured relationships using primary and foreign keys, ensuring data consistency and normalization. The database schemas have been designed in a normalized form to eliminate redundancy and ensure consistency of data. This enhances system performance and ease of data management.

The system includes:

- Profile management
- Scholarship recommendations
- Mentorship booking
- Deadline tracking
- Application management
- Review system

The system is built using Visual Studio Code, XAMPP and MySQL.

The system has several input forms that aim to enhance usability and human-computer interaction through easy and efficient data entry. For instance, the user can quickly select continent and university via dropdown menus, minimising input and processing time. Date inputs have calendars for selecting dates, while GPA is entered using up and down buttons. Rates are input via a five-star menu for quick selection. This design adheres to the principles of Human-Computer Interaction (HCI) to improve user experience and reduce errors.

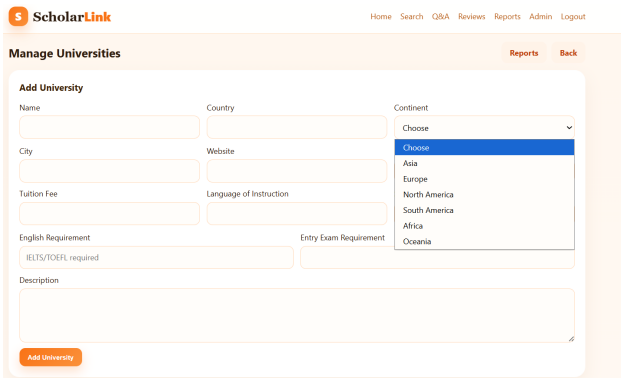


Fig. 5. Input Dropdown

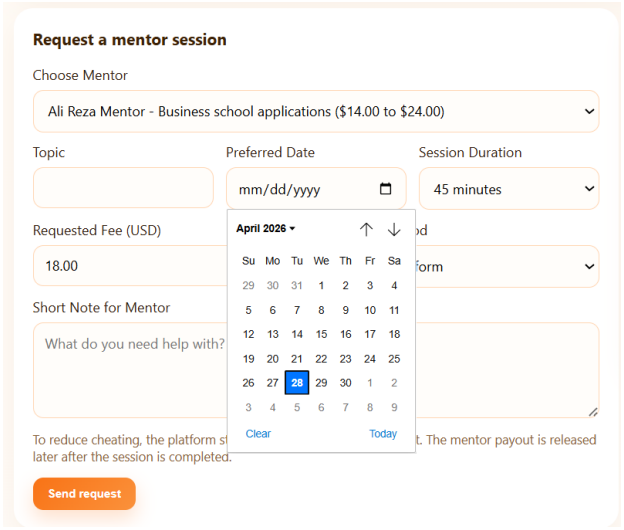


Fig. 6. Input Calendar

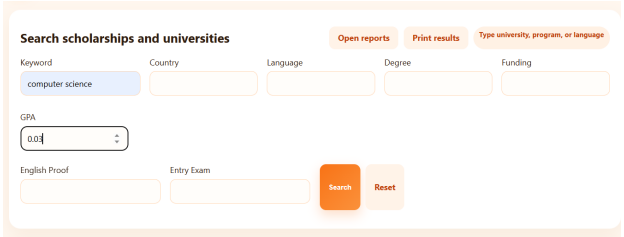


Fig. 7. Input GPA Control

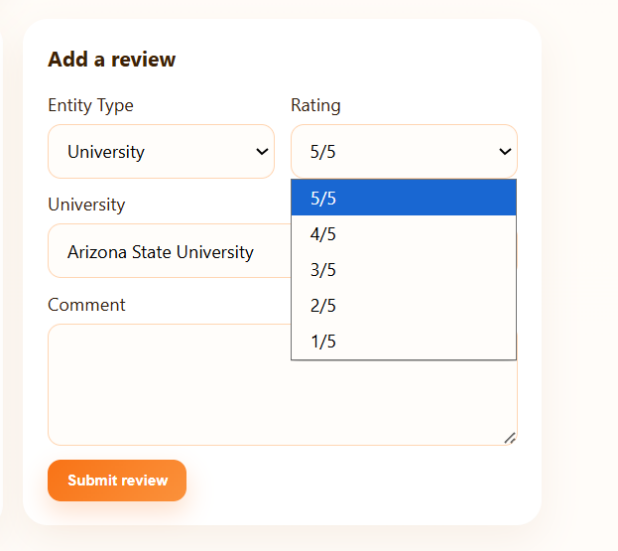


Fig. 8. Input Rating

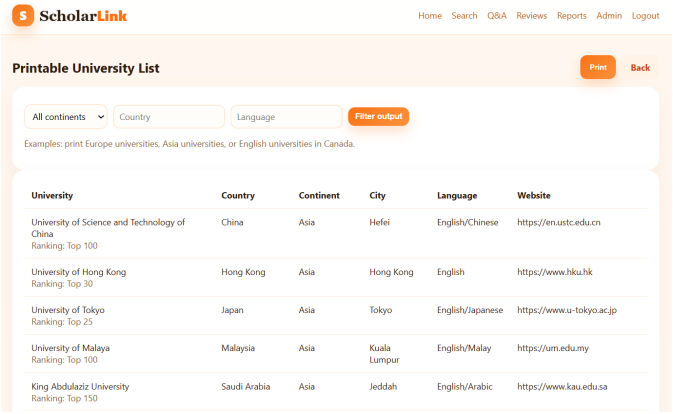


Fig. 10. University List Report

V. Result Analysis

The system provides output reports, which can be printed and presented in the form of charts and graphs. The reports allow data analysis of the system. The filtering and search feature enhances the search for universities and scholarships by eliminating irrelevant search results and providing time efficiency for the users. The reports have advantages like effective decision-making, data presentation and system monitoring. Further, a leaderboard is integrated into the system, which showcases the main users or activities.

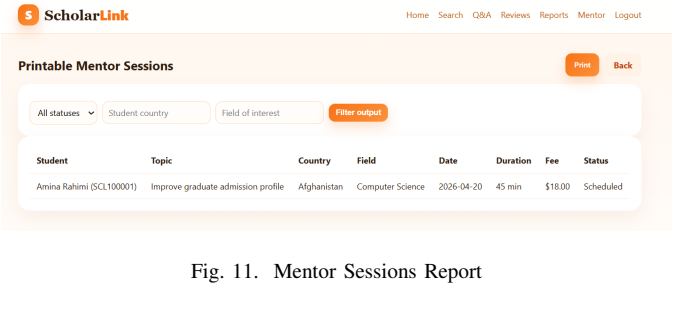


Fig. 11. Mentor Sessions Report

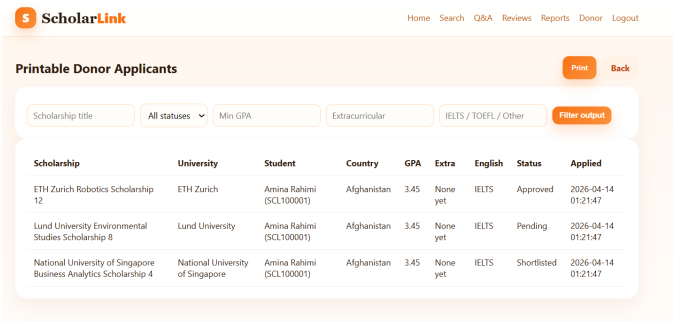


Fig. 9. Donor Report

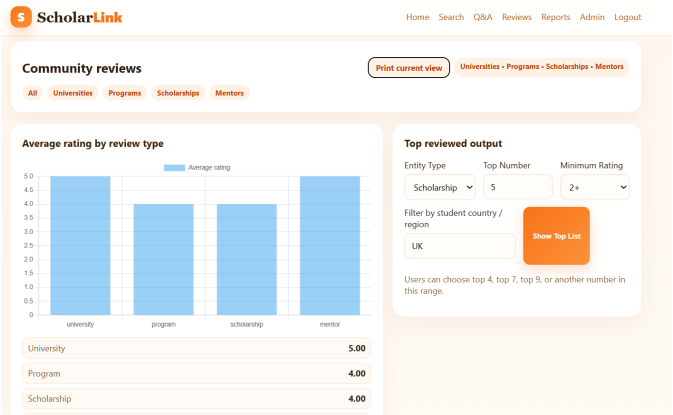


Fig. 12. Reviews Report

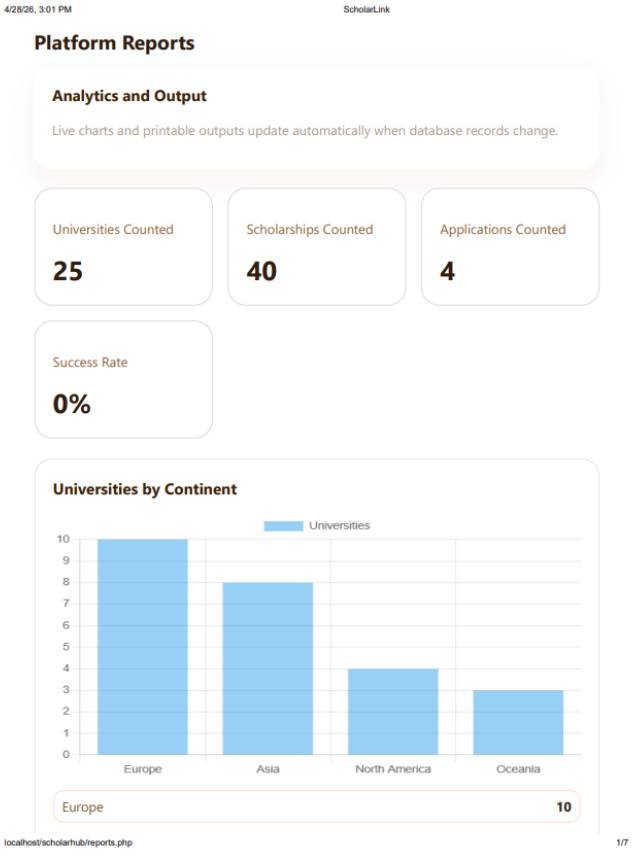


Fig. 13. General Report Output

VI. Conclusion

This paper presented Scholar Link, a web-based system to aid the search for scholarships, universities or programs. The system helps organise the data, provides feedback and reminders address some of the limitations of existing systems. The system reduces information overload, facilitates decision-making and enhances user trust in the content with credible and organised information. This is particularly beneficial for those who need scholarships to meet their academic aspirations. Future research might include more advanced recommendation features, live data feeds and apps for mobile devices. improve the system.

References

[1] E. Herbaut and K. Geven, 2019. [2] N. Kamal et al., IEEE Access, 2024. [3] R. Li et al., Behavioral Sciences, 2023. [4] P. Mabotha and B. Ngcamu, 2026. [5] M. Serwornoo et al., 2024.